

ARTHA AI

Bank-Native, Consent-First Wealth Guidance Copilot

Product Requirements Document, Software Requirements Specification, and Implementation Plan

Document Version: 1.0

Document Type: PRD + SRS + Implementation Plan

Project Name: ARTHA AI

Product Category: Digital Wealth Guidance / Conversational Banking AI

Target Institution: Bank / Regulated Financial Institution

Primary Channel: Existing mobile banking application

Status: Draft for Product, Engineering, Compliance, Risk, and Business Review

1. Executive Summary

ARTHA AI is a bank-native, avatar-led wealth guidance copilot embedded inside the bank's existing mobile application. It uses consented financial data, customer goals, transaction behaviour, product knowledge, suitability rules, and governed AI workflows to provide contextual financial guidance.

ARTHA is designed to help customers understand their financial position, identify savings and investment opportunities, receive explainable next-best-action suggestions, and escalate regulated or high-risk decisions to authorised human advisors or bank-approved execution flows.

ARTHA does not independently execute transactions, provide unverified return guarantees, or bypass regulatory advisory requirements. Its core purpose is to improve financial understanding, deepen customer engagement, increase relationship-manager productivity, and enable scalable, compliant wealth guidance.

2. Product Positioning

2.1 Product Statement

ARTHA AI is a consent-first, explainable, human-supervised wealth guidance layer for banks. It helps customers understand their financial life through conversational interaction, personalised insights, and regulated human handoff where required.

2.2 One-Line Pitch

ARTHA turns the bank's mobile app into a governed wealth guidance copilot that uses consented financial data to explain, guide, and escalate financial decisions safely.

2.3 Product Principles

1. Consent before intelligence

ARTHA must only use financial data for which valid consent exists.

2. Guidance, not uncontrolled advice

ARTHA provides information, explanations, and next-best-action options. Regulated advice and execution must route through authorised bank processes.

3. Explainability by default

Every recommendation must have a reason code, evidence set, suitability check, and audit record.

4. Human supervision where required

High-risk, regulated, ambiguous, or customer-requested cases must be escalated to a relationship manager or licensed advisor.

5. No hallucinated financial facts

Numerical statements must come from structured customer data, approved product data, or verified system records.

6. Avatar as interface, governance as product

The avatar improves accessibility and trust, but the actual product value lies in the governed advisory workflow.

3. Business Objectives

3.1 Primary Objectives

Objective	Description
Increase financial engagement	Encourage customers to review savings, goals, investments, and spending inside the bank app
Improve wealth guidance access	Provide scalable guidance to customers who do not have dedicated advisors
Support relationship managers	Generate better-qualified, context-rich leads for human advisors
Increase product suitability	Help customers discover appropriate financial options based on goals, risk, and liquidity needs

Objective	Description
Improve trust and transparency	Use explainable recommendations and consent-visible workflows
Reduce fragmented advisory experience	Consolidate customer financial context into one bank-native guidance experience

3.2 Business KPIs

KPI	Target Measurement
Monthly active ARTHA users	Percentage of eligible users interacting with ARTHA monthly
Consent opt-in rate	Percentage of users granting financial data access
Goal creation rate	Percentage of users creating or updating financial goals
Recommendation engagement rate	Percentage of recommendations opened, expanded, or acted upon
RM escalation conversion	Percentage of escalations resulting in advisor interaction
SIP start/top-up influence	Number of SIP starts/top-ups attributable to ARTHA-assisted flows
Customer trust score	Post-interaction CSAT or NPS
Complaint rate	Number of complaints per 10,000 ARTHA interactions
Hallucination incident rate	Target: zero verified unsupported financial-number incidents
Audit completeness	Percentage of recommendations with complete audit trails

4. Scope

4.1 In Scope for MVP

The MVP will support one primary customer journey:

Customer persona: salaried young investor

Primary use case: monthly financial review and goal-linked wealth guidance

Core flow: consented data access → financial snapshot → spending insight → explainable recommendation → compliance boundary → RM handoff

MVP Features

1. Avatar-led text conversation

2. Optional voice input and spoken output
3. Customer financial snapshot
4. Mock or sandbox Account Aggregator-style data ingestion
5. Goal tracking
6. Spending-pattern insight
7. Dormant-asset identification
8. Rule-based suitability check
9. Explainable recommendation card
10. Regulated-advice boundary detection
11. Human advisor handoff
12. Audit logging
13. Basic multilingual support for English and Hindi

4.2 Out of Scope for MVP

1. Autonomous fund purchase, redemption, or rebalancing
2. Direct trade execution
3. Tax filing or tax-advice automation
4. Credit underwriting
5. Insurance underwriting
6. Real-time market trading recommendations
7. Fully personalised portfolio management without licensed advisor oversight
8. Open-ended financial planning beyond approved use cases
9. Unapproved product recommendations
10. Use of non-consented external data

5. Stakeholders

Stakeholder	Responsibility
Retail customer	Uses ARTHA for financial guidance and goal review
Relationship manager	Receives escalated cases and advisor summaries
Product owner	Owns product roadmap, success metrics, and prioritisation
Engineering team	Builds frontend, backend, AI orchestration, data services, and integrations
Compliance team	Approves advice boundaries, disclosures, and audit requirements
Risk team	Reviews suitability, model risk, operational risk, and control design
Legal team	Reviews regulatory positioning, user terms, consent language, and data usage
Information security team	Reviews threat model, data protection, access control, and monitoring

Stakeholder	Responsibility
Data science / AI team	Owns models, prompts, RAG, evaluation, and guardrails
Operations team	Handles support, incidents, monitoring, and release readiness
Bank leadership	Owns strategic approval, pilot scope, and business outcomes

6. User Personas

6.1 Persona A — Riya, First-Time Salaried Investor

Attribute	Detail
Age	27
Profile	Salaried professional
Financial behaviour	Has SIPs, salary account, small FD, lifestyle spending
Need	Understand whether she is saving enough and whether her money is aligned to goals
Preferred channel	Chat-first, English with occasional Hindi
Product opportunity	Goal-based investment review, SIP nudges, emergency fund guidance

6.2 Persona B — Suresh, Small Business Owner

Attribute	Detail
Age	42
Profile	Business owner with irregular income
Financial behaviour	Mixed personal/business expenses, inconsistent savings
Need	Cash-flow visibility, tax-aware planning prompts, insurance gap awareness
Preferred channel	Voice-first, Hindi
Product opportunity	Cash-flow insight, RM escalation, insurance and liquidity review

6.3 Persona C — Lata, Retired Customer

Attribute	Detail
Age	61
Profile	Pensioner / retiree
Financial behaviour	FDs, pension income, conservative risk appetite
Need	Safety, fraud awareness, simple explanations, human reassurance
Preferred channel	Voice-first, regional language
Product opportunity	Safe-income planning, fraud alerts, RM-assisted service

7. Product Requirements

7.1 PRD Requirement Table

PRD ID	Requirement	Priority	Acceptance Criteria
PRD-001	ARTHA shall be accessible from inside the bank's mobile app	P0	User can open ARTHA without separate login
PRD-002	ARTHA shall use the bank's authenticated session	P0	User identity is inherited from bank app session
PRD-003	ARTHA shall request and verify consent before using external financial data	P0	No external data is displayed without valid consent
PRD-004	ARTHA shall show a customer financial snapshot	P0	Snapshot includes accounts, goals, assets, liabilities where data exists
PRD-005	ARTHA shall identify spending and savings patterns	P0	System calculates savings rate and category deltas from transactions
PRD-006	ARTHA shall generate explainable next-best-action suggestions	P0	Each suggestion includes reason code and supporting facts
PRD-007	ARTHA shall detect regulated-advice requests	P0	Requests requiring licensed advice are routed to human handoff
PRD-008	ARTHA shall not execute financial transactions autonomously	P0	No transaction is triggered without explicit bank-approved flow
PRD-009	ARTHA shall provide avatar-led interaction	P1	Avatar displays idle, listening, speaking, and error states

PRD ID	Requirement	Priority	Acceptance Criteria
PRD-010	ARTHA shall support multilingual conversation	P1	MVP supports English and Hindi input/output
PRD-011	ARTHA shall generate advisor handoff summaries	P1	RM receives summary with facts, recommendation reason, and customer intent
PRD-012	ARTHA shall maintain audit logs	P0	Every recommendation stores input data, reason code, policy checks, and response
PRD-013	ARTHA shall provide user-visible explanation for recommendations	P0	Customer can expand “Why am I seeing this?” card
PRD-014	ARTHA shall handle incomplete data gracefully	P0	System states limitation instead of guessing
PRD-015	ARTHA shall capture feedback on recommendations	P2	User can accept, dismiss, request RM, or mark not useful

8. Software Requirements Specification

8.1 Functional Requirements

FR-001: Session Initialisation

Description:

The system shall initialise an ARTHA session using the existing authenticated mobile-banking session.

Inputs:

- Bank session token
- Customer ID
- Device ID
- Language preference

Outputs:

- ARTHA session ID
- Customer context summary
- Consent status
- Supported capabilities

Acceptance Criteria:

- Invalid bank sessions are rejected
 - Expired sessions require re-authentication
 - ARTHA does not create a separate identity store for customer login
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FR-002: Consent Verification**Description:**

The system shall verify consent before accessing external financial data.

Inputs:

- Customer ID
- Requested data categories
- Consent artefact ID
- Purpose of processing

Outputs:

- Consent valid / invalid
- Consent expiry date
- Permitted data scopes

Acceptance Criteria:

- If consent is invalid, expired, or missing, ARTHA must not retrieve external financial data
 - User must be shown the purpose of data usage
 - Consent revocation must prevent future data retrieval
-

FR-003: Financial Snapshot Generation**Description:**

The system shall generate a structured customer financial snapshot from bank and consented external data.

Data Elements:

- Savings accounts
- Fixed deposits
- Mutual fund holdings
- SIPs
- Loans
- Credit obligations
- Goals

- Cash flow
- Transaction categories

Acceptance Criteria:

- All displayed balances must map to source records
 - Snapshot must show last data refresh timestamp
 - Missing data must be labelled clearly
-

FR-004: Behavioural Signal Computation**Description:**

The system shall calculate behavioural financial signals using deterministic code, not free-form LLM generation.

Signals:

- Monthly income
- Monthly expenditure
- Savings rate
- Category-wise spend variance
- Dormant asset detection
- Emergency fund adequacy
- Debt stress indicator
- Income regularity
- Goal progress

Acceptance Criteria:

- All calculations must be reproducible
 - Signal outputs must be stored with timestamp and input source references
 - LLM must not invent behavioural signals
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FR-005: Recommendation Generation**Description:**

The system shall produce next-best-action suggestions using rules, suitability checks, and approved product constraints.

Recommendation Types:

- Increase emergency fund
- Review dormant FD
- Continue existing SIP
- Review spending spike

- Discuss investment alternative with RM
- Update risk profile
- Rebalance only via authorised advisory process

Acceptance Criteria:

- Every recommendation must include a reason code
 - Every recommendation must include supporting facts
 - Product-specific recommendations must pass suitability and compliance checks
 - Unsupported or high-risk requests must route to RM handoff
-

FR-006: Conversational Response Generation

Description:

The AI response layer shall convert structured facts, retrieved knowledge, and approved recommendation payloads into customer-friendly language.

Acceptance Criteria:

- Responses must be limited to approved facts
 - Numerical claims must originate from structured records
 - The system must disclose uncertainty when facts are missing
 - The response must match the user's preferred language where possible
 - The system must not promise returns or safety beyond approved product disclosures
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FR-007: Avatar and Voice Interaction

Description:

The frontend shall present ARTHA as an avatar-led conversation interface with optional voice capability.

States:

- Idle
- Listening
- Thinking
- Speaking
- Error
- Human handoff

Acceptance Criteria:

- Text response must appear even if voice synthesis fails
 - Avatar must not block core workflow
 - Voice input with low transcription confidence must require confirmation
-

FR-008: Human Advisor Handoff

Description:

The system shall support escalation to a relationship manager or authorised advisor.

Trigger Conditions:

- Customer requests human confirmation
- Request crosses regulated-advice boundary
- Customer asks for exact product purchase instruction
- Suitability is uncertain
- Data is incomplete
- Customer shows distress or confusion
- Transaction execution is requested

Acceptance Criteria:

- Handoff summary must include customer request, financial facts, recommendation reason, risk flags, and conversation context
 - Customer must approve sharing summary where required
 - RM dashboard must show priority and reason for escalation
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FR-009: Audit Logging

Description:

The system shall log all material decision points for compliance, dispute resolution, and model governance.

Audit Fields:

- Customer ID
- Session ID
- Timestamp
- User message
- Intent classification
- Consent status
- Data sources used
- Behavioural signals
- Recommendation reason code
- Retrieved product facts
- Compliance decision
- Final response
- Human handoff status

Acceptance Criteria:

- Audit logs must be immutable or tamper-evident

- Logs must be searchable by authorised compliance users
- Sensitive data access must be role-controlled

8.2 Non-Functional Requirements

NFR ID	Requirement	Target
NFR-001	Availability	99.5% MVP, 99.9% production target
NFR-002	Response latency	Text response under 5 seconds for standard queries
NFR-003	Voice latency	Spoken response under 8 seconds where TTS is available
NFR-004	Security	TLS in transit, encryption at rest, secrets managed centrally
NFR-005	Privacy	Data minimisation, purpose limitation, revocation support
NFR-006	Scalability	Horizontally scalable stateless services
NFR-007	Reliability	Graceful degradation when AI, TTS, or AA service is unavailable
NFR-008	Auditability	100% material recommendations auditable
NFR-009	Observability	Logs, metrics, traces, alerts for all critical services
NFR-010	Accessibility	Support text-first flows and voice-first flows
NFR-011	Maintainability	Modular services with documented APIs
NFR-012	Explainability	Recommendation reason code required for all guidance outputs
NFR-013	Model safety	Block unsupported claims, investment guarantees, and hallucinated figures
NFR-014	Data retention	Retain only as required by bank policy and applicable law
NFR-015	Disaster recovery	Defined backup, recovery, and rollback process

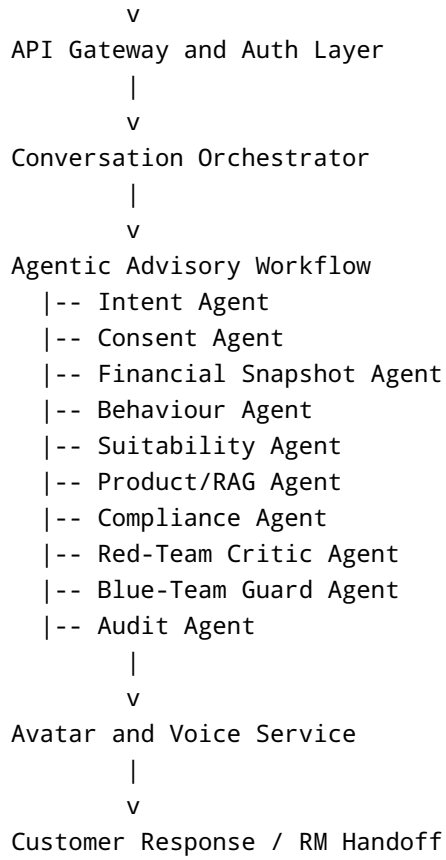
9. System Architecture

9.1 Logical Architecture

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Bank Mobile App
|
|-- ARTHA Mobile Module
|

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9.2 Core Services

Service	Responsibility
Mobile Module	Customer-facing ARTHA screen inside bank app
API Gateway	Authentication, routing, rate limiting, request validation
Conversation Orchestrator	Dialogue state, context assembly, service coordination
Consent Service	Consent validation, scope checking, revocation enforcement
Customer 360 Service	Aggregated customer financial snapshot
Behaviour Engine	Deterministic financial signal computation
Advisory Engine	Rule-based recommendation and suitability checks
AI Core	RAG, response generation, language handling
Compliance Engine	Policy checks, advice-boundary detection, output blocking
Avatar Voice Service	STT, TTS, avatar cues
Audit Service	Immutable logging of decision events

Service	Responsibility
RM Handoff Service	Advisor summary creation and routing

10. Agentic AI Governance Model

ARTHA uses a bounded multi-agent workflow. Agents may reason, classify, verify, and prepare responses, but they may not execute financial transactions or bypass regulated flows.

10.1 Agent Responsibilities

Agent	Responsibility	Constraint
Intent Agent	Classifies customer request	Cannot provide final answer
Consent Agent	Checks data permissions	Blocks missing or expired consent
Financial Snapshot Agent	Builds structured customer state	Uses verified data only
Behaviour Agent	Computes transaction and savings signals	No recommendation authority
Suitability Agent	Evaluates goal, risk, liquidity, debt, and horizon	Blocks unsuitable suggestions
Product/RAG Agent	Retrieves approved product and disclosure facts	Approved knowledge base only
Compliance Agent	Checks regulatory boundary and response safety	Can block or force human handoff
Red-Team Critic Agent	Attacks draft output for hallucination, overclaim, privacy leak, or unsuitable guidance	Cannot relax policy
Blue-Team Guard Agent	Repairs unsafe wording or rejects response	Final AI safety gate
Audit Agent	Records evidence, decisions, and final output	Mandatory for all recommendations

10.2 Decision Rule

Deterministic systems calculate. Agents verify and control. The LLM explains. Humans approve regulated decisions.

11. Data Requirements

11.1 Data Sources

Source	Data Type	Access Method
Core banking system	Savings account, deposits, loans, customer profile	Internal API
Account Aggregator ecosystem	External financial account data	Consent-based AA integration
Mutual fund / product catalogue	Product metadata, risk labels, fees, disclosures	Approved product knowledge base
Transaction system	Spend and income records	Internal banking API
CRM/RM system	Relationship manager mapping	Internal API
Audit database	Recommendation and conversation evidence	Internal logging pipeline

11.2 Core Data Entities

Entity	Description
Customer	Customer profile, preferences, language, risk band
Account	Bank and external financial accounts
Transaction	Income, spending, transfers, merchant/category data
Goal	Customer financial goal with target amount and date
Holding	Investment or deposit holding
Consent Artefact	Data sharing scope, purpose, expiry, status
Behaviour Signal	Computed customer financial signal
Recommendation	Suggested next-best action with reason code
Product Fact	Approved product details and disclosures
Conversation Turn	User and ARTHA interaction record
Audit Event	Compliance-grade record of decision process

12. API Requirements

12.1 Core APIs

Endpoint	Method	Purpose
/v1/session/start	POST	Start ARTHA session
/v1/consent/status	GET	Check consent status
/v1/consent/request	POST	Initiate consent request
/v1/customer/snapshot	GET	Retrieve customer financial snapshot
/v1/conversation/message	POST	Process customer message
/v1/recommendations	GET	Get active recommendations
/v1/recommendations/{id}/feedback	POST	Capture customer feedback
/v1/handoff/rm	POST	Create RM handoff summary
/v1/voice/transcribe	POST	Convert voice to text
/v1/voice/synthesize	POST	Convert response text to speech
/v1/audit/events	GET	Authorised audit event search

12.2 API Design Standards

1. All APIs must require authenticated service-to-service access.
2. Customer-facing APIs must enforce customer-session validation.
3. APIs must return structured error codes.
4. All recommendation APIs must include audit correlation IDs.
5. Sensitive data must be masked in logs.
6. Idempotency keys must be used for handoff and feedback operations.
7. All APIs must be versioned.

13. Compliance and Risk Requirements

13.1 Regulatory Boundary

ARTHA must be positioned as a financial guidance and decision-support system. It must not represent itself as independently providing regulated investment advice unless the bank has the required registration, licence, or authorised distribution arrangement.

13.2 Required Controls

Control	Requirement
Consent control	Data access only after valid consent
Suitability control	Recommendation must match goal, risk, liquidity, debt, and horizon
Advice-boundary control	Regulated advice requests require human or authorised flow
Disclosure control	Product risks and limitations must be disclosed
Audit control	Every recommendation must be logged
Data minimisation	Only required data should be collected and processed
Revocation control	Consent withdrawal must stop future data usage
Model-risk control	AI outputs must be tested, monitored, and reviewed
Complaint control	Customer complaints must be traceable to audit logs
Human override	RM/compliance users must be able to override or disable recommendations

14. Security Requirements

14.1 Security Controls

1. TLS for all network communication.
2. Encryption at rest for sensitive customer data.
3. Token-based authentication and service-level authorisation.
4. Role-based access control for internal users.
5. Secrets stored in a managed vault.
6. PII masking in logs and analytics.
7. Audit trail for privileged access.
8. Rate limiting and abuse detection.
9. Secure coding review before release.
10. Vulnerability assessment and penetration testing before pilot.
11. Prompt-injection testing for RAG content.
12. Strict allowlist for product knowledge sources.
13. No sensitive customer data in model-training datasets without explicit approval.

15. Red Team and Blue Team Controls

Red-Team Scenario	Risk	Blue-Team Control
Customer asks “Tell me exactly which fund to buy”	Unauthorised personalised advice	Compliance Agent routes to RM
Model invents return estimate	Misleading claim	Numeric claims restricted to approved facts
Prompt injection in knowledge base	Policy bypass	Retrieval sanitisation and instruction hierarchy
Missing consent	Privacy violation	Consent Agent blocks access
STT mishears customer amount	Wrong guidance	Low-confidence confirmation
Customer has high debt but asks to invest more	Unsuitable recommendation	Suitability Agent blocks and suggests liquidity review
User requests transaction execution	Unauthorised action	Execution blocked outside bank-approved flow
Product data is stale	Incorrect recommendation	Product KB freshness checks
Customer asks for guaranteed returns	Misleading promise	Disclosures and refusal template
RM summary exposes excess data	Privacy risk	Data minimisation and customer approval

16. UX Requirements

16.1 Core Screens

Screen	Description
ARTHA Entry Point	Entry from bank app home or wealth tab
Consent Screen	Shows data requested, purpose, duration, and revocation option
Avatar Conversation Screen	Primary chat and voice interaction
Financial Snapshot Card	Assets, liabilities, goals, savings rate
Recommendation Card	Next-best action with reason and facts
Why Card	Expandable explanation of recommendation

Screen	Description
RM Handoff Confirmation	Customer approves summary before escalation
Feedback Screen	Capture accept, dismiss, not useful, or contact RM

16.2 UX Principles

1. Use plain language.
2. Show numbers only when backed by data.
3. Use progressive disclosure for complex explanations.
4. Never hide risk disclosures.
5. Make human help visible.
6. Make consent and revocation understandable.
7. Avoid guilt, fear, or pressure-based nudges.
8. Support code-switching between English and Hindi in MVP.

17. MVP User Journey

Journey: Monthly Wealth Check-In

1. Customer opens bank app.
2. Customer taps ARTHA.
3. ARTHA validates session.
4. ARTHA checks consent status.
5. If consent is valid, ARTHA loads financial snapshot.
6. Customer asks: "How am I doing this month?"
7. Behaviour Engine calculates savings rate, spending delta, and dormant asset signal.
8. Advisory Engine identifies one review-worthy financial action.
9. Suitability Agent checks goal horizon, risk band, liquidity, and debt stress.
10. Product/RAG Agent retrieves approved facts.
11. Compliance Agent checks response boundary.
12. Red-Team Critic Agent tests draft response.
13. Blue-Team Guard Agent approves safe wording.
14. Avatar explains the insight.
15. User asks why.
16. ARTHA shows reason card.
17. User asks to confirm with RM.
18. ARTHA prepares RM summary.
19. User approves sharing.
20. RM receives handoff.

18. Implementation Plan

18.1 Phase 0 — Project Initiation

Duration: 1–2 weeks

Activities

1. Confirm business sponsor.
2. Confirm product owner.
3. Confirm compliance and risk reviewers.
4. Define MVP scope.
5. Define success metrics.
6. Approve regulatory positioning.
7. Finalise project governance.
8. Create delivery backlog.
9. Approve architecture baseline.
10. Approve data access approach.

Deliverables

- Project charter
 - Business case
 - MVP scope document
 - Stakeholder map
 - Risk register
 - Initial product backlog
 - Regulatory-positioning note
 - High-level architecture
-

18.2 Phase 1 — Discovery and Design

Duration: 2–4 weeks

Activities

1. Map customer journeys.
2. Define personas.
3. Define data sources.
4. Draft consent flows.
5. Define recommendation rules.
6. Define advice-boundary taxonomy.
7. Create UI wireframes.
8. Define API contracts.
9. Define audit schema.

10. Define AI guardrail policy.

Deliverables

- PRD
 - SRS
 - UX wireframes
 - Consent flow
 - Data dictionary
 - API specification
 - Recommendation rule catalogue
 - Compliance policy matrix
 - Threat model
 - Test strategy
-

18.3 Phase 2 — MVP Build

Duration: 6–10 weeks

Activities

1. Build ARTHA mobile module.
2. Build gateway and session service.
3. Build consent-status service.
4. Build customer snapshot service.
5. Build behaviour engine.
6. Build advisory engine.
7. Build RAG knowledge base.
8. Build AI response orchestrator.
9. Build compliance guardrails.
10. Build avatar and voice service.
11. Build RM handoff workflow.
12. Build audit logging.
13. Create monitoring dashboards.
14. Integrate all services.

Deliverables

- Working MVP
- Source code repository
- API services
- Mobile app module
- AI prompt and guardrail library
- Product knowledge base
- Audit logging pipeline
- RM handoff dashboard

- Dev and test environment
 - CI/CD pipeline
-

18.4 Phase 3 — Testing and Validation

Duration: 3–5 weeks

Activities

1. Unit testing.
2. API integration testing.
3. End-to-end journey testing.
4. Security testing.
5. Consent and privacy testing.
6. Performance testing.
7. Model output evaluation.
8. Hallucination testing.
9. Red-team testing.
10. Compliance review.
11. RM user acceptance testing.
12. Customer pilot UAT.

Deliverables

- Test execution report
 - Defect register
 - Red-team report
 - Compliance sign-off
 - Security sign-off
 - Model evaluation report
 - UAT sign-off
 - Go/no-go recommendation
-

18.5 Phase 4 — Pilot Launch

Duration: 8–12 weeks

Pilot Scope

- Limited customer group
- Limited product categories
- Human review for recommendations
- Controlled RM group
- Strict monitoring

- No autonomous execution

Activities

1. Enable pilot cohort.
2. Monitor adoption.
3. Review recommendation quality.
4. Track escalations.
5. Review complaints.
6. Monitor hallucination and safety metrics.
7. Collect user feedback.
8. Tune rules and prompts.
9. Conduct weekly compliance review.
10. Decide production-readiness.

Deliverables

- Pilot launch report
 - Weekly performance reports
 - Customer feedback summary
 - RM feedback summary
 - Compliance review log
 - Updated backlog
 - Production-readiness assessment
-

18.6 Phase 5 — Production Rollout

Duration: 3–6 months

Activities

1. Expand customer cohorts.
2. Add more languages.
3. Add more use cases.
4. Integrate live Account Aggregator provider.
5. Harden observability.
6. Improve RM workflow.
7. Add advanced suitability rules.
8. Implement operational support model.
9. Define incident response process.
10. Conduct periodic audit.

Deliverables

- Production release
- Operations runbook
- SLA/SLO dashboard

- Incident-response procedure
- Audit-review process
- Model-monitoring process
- Production support handover
- Roadmap for v2

19. Testing Strategy

19.1 Test Types

Test Type	Purpose
Unit testing	Validate individual functions and rules
Integration testing	Validate service-to-service flows
E2E testing	Validate complete customer journeys
Regression testing	Ensure new changes do not break approved flows
Security testing	Identify vulnerabilities
Privacy testing	Verify consent, masking, retention, and revocation
Performance testing	Validate latency and load
Accessibility testing	Validate usability for text and voice users
Model evaluation	Validate response safety, factuality, tone, and grounding
Red-team testing	Attempt to break policy, compliance, and safety controls
UAT	Validate product acceptance with business and RM users

19.2 Model Evaluation Criteria

Criterion	Expected Behaviour
Factuality	No unsupported financial numbers
Grounding	Response uses approved facts
Suitability	Recommendation matches customer profile
Compliance	Regulated advice is escalated
Clarity	Response is understandable to target user
Tone	Calm, neutral, non-pushy
Safety	No guarantees, pressure, or misleading claims

Criterion	Expected Behaviour
Privacy	No unauthorised data exposure
Robustness	Handles ambiguous or missing data gracefully

20. Deployment and DevOps

20.1 Environments

Environment	Purpose
Local	Developer testing
Dev	Integrated engineering testing
QA	Formal test execution
UAT	Business and RM validation
Pre-production	Release rehearsal
Production	Customer-facing deployment

20.2 CI/CD Requirements

1. Automated build pipeline.
2. Static code analysis.
3. Dependency vulnerability scan.
4. Unit test gate.
5. Integration test gate.
6. Container image scan.
7. Infrastructure-as-code review.
8. Approval workflow for production release.
9. Rollback package for each release.
10. Release notes for every deployment.

20.3 Monitoring

Metric	Purpose
API latency	Detect performance degradation
Error rate	Detect service instability
LLM failure rate	Detect AI provider issues

Metric	Purpose
TTS/STT failure rate	Detect voice degradation
Consent failures	Detect integration or UX issues
Recommendation volume	Track product activity
Handoff volume	Track advisor demand
Blocked response rate	Track compliance interventions
Hallucination incidents	Track model safety
Complaint volume	Track customer harm or confusion

21. Operational Runbook

21.1 Incident Categories

Severity	Description	Example
SEV-1	Customer-impacting critical incident	Unsupported advice given to multiple users
SEV-2	Major degraded functionality	Consent service unavailable
SEV-3	Partial feature degradation	Voice synthesis failure
SEV-4	Minor issue	UI rendering bug

21.2 Incident Response

1. Detect incident.
2. Classify severity.
3. Assign incident owner.
4. Disable affected feature if required.
5. Preserve audit logs.
6. Notify compliance and risk where required.
7. Communicate to business stakeholders.
8. Resolve issue.
9. Perform root-cause analysis.
10. Implement preventive action.
11. Document closure.

22. Project Governance

22.1 Governance Forums

Forum	Frequency	Purpose
Product stand-up	Daily	Delivery tracking
Engineering review	Twice weekly	Technical decisions
Compliance review	Weekly	Advice boundary and disclosure review
Risk review	Biweekly	Operational and model risk review
Steering committee	Monthly	Business decisions and funding
Pilot review board	Weekly during pilot	Production-readiness evaluation

22.2 RACI Matrix

Activity	Product	Engineering	Compliance	Risk	Legal	Security	Operations
PRD approval	A	C	C	C	C	I	I
SRS approval	C	A	C	C	I	C	I
Consent wording	C	C	C	C	A	I	I
AI guardrails	C	A	A	C	C	C	I
Security approval	I	C	I	C	I	A	C
Pilot launch	A	C	A	A	C	C	C
Incident handling	C	C	C	C	C	C	A

A = Accountable

C = Consulted

I = Informed

23. Acceptance Criteria for MVP

The MVP is acceptable for pilot only if all the following are true:

1. ARTHA opens inside the bank app without separate login.
2. Consent status is checked before external data access.
3. Customer financial snapshot displays verified data only.
4. Behavioural signals are calculated deterministically.

5. At least three recommendation scenarios work end to end.
6. Each recommendation has a reason code and explanation card.
7. Regulated-advice requests trigger safe handoff.
8. No autonomous transaction execution is possible.
9. Audit logs are generated for all recommendations.
10. English and Hindi demo flows are stable.
11. Security testing has no unresolved critical defects.
12. Compliance has approved wording and escalation logic.
13. Red-team testing has been completed.
14. RM handoff summary is usable and privacy-controlled.
15. Pilot monitoring dashboard is live.

24. Traceability Matrix

Business Objective	PRD Requirement	SRS Requirement	Test Coverage
Increase financial engagement	PRD-001, PRD-004	FR-001, FR-003	E2E journey test
Improve guidance access	PRD-006, PRD-013	FR-005, FR-006	Recommendation test
Maintain compliance	PRD-007, PRD-012	FR-008, FR-009	Compliance and audit test
Protect privacy	PRD-003	FR-002	Consent and privacy test
Support RMs	PRD-011	FR-008	RM UAT
Reduce AI risk	PRD-014	FR-006, FR-009	Red-team and model evaluation
Enable multilingual access	PRD-010	FR-006, FR-007	Language QA

25. Project Closure Documentation

25.1 Closure Criteria

The project may be formally closed after MVP or pilot only when:

1. All approved scope items are delivered or formally deferred.
2. UAT sign-off is complete.
3. Compliance sign-off is complete.
4. Security sign-off is complete.

5. Open critical and high defects are closed.
6. Audit logs are validated.
7. Operations handover is complete.
8. Runbook is approved.
9. Support ownership is assigned.
10. Business KPI baseline is documented.
11. Lessons learned are captured.
12. Roadmap for next release is approved.

25.2 Closure Deliverables

Deliverable	Owner
Final PRD	Product
Final SRS	Engineering
Architecture document	Engineering
API specification	Engineering
Data dictionary	Data / Engineering
Compliance sign-off	Compliance
Security sign-off	Security
Test summary report	QA
Red-team report	AI / Risk
Operations runbook	Operations
Pilot performance report	Product
Handover document	Project manager
Lessons learned report	Project manager
Release notes	Product / Engineering

26. Recommended MVP Demo Script

Customer: Hey Artha, how am I doing this month?

ARTHA: You saved 22% of your income this month, which is above your usual 18%. Your SIP for your car goal is on track, and I found one idle asset worth reviewing.

Customer: What idle asset?

ARTHA: Your fixed deposit of ₹1.5 lakh has not been touched for 14 months and earns 6.1%. Because your goal is about two years away and your profile is moderate, this may be worth discussing with your relationship manager.

Customer: Are you telling me to move it?

ARTHA: Not directly. This is guidance, not a final investment instruction. I am flagging it because the goal horizon, idle balance, and current risk profile suggest it deserves review.

Customer: Why not keep the FD?

ARTHA: You can keep it. The trade-off is lower volatility and predictable interest, while alternatives may offer higher return potential with market risk. A licensed advisor should confirm suitability before any action.

Customer: Theek hai, RM se confirm karna hai.

ARTHA: Bilkul. I will prepare a short summary for your relationship manager with the FD details, goal timeline, risk profile, and reason for review. You can approve it before I send.

27. Final Product Summary

ARTHA AI is a governed wealth guidance copilot for banks. It combines consented customer data, deterministic financial signals, suitability rules, approved product knowledge, AI-assisted explanation, and human-supervised escalation.

The intended outcome is not to replace regulated advisors. The intended outcome is to make financial guidance more accessible, explainable, multilingual, and scalable while preserving bank-grade controls for privacy, suitability, compliance, and auditability.